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Batch: June

Course: Data Analytics with AI

Assignment: SQL Basics Assignment

Submission Date:- 29-07-2026

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➔ Write the Question and Answers.

Q1. Create a New Database and Table for Employees Task: Create a new database named company_db and Create a table named employees with the following columns:

Column	Data Type	Constraint
employee_id	INT	PRIMARY KEY
first_name	VERCHAR(50)	
last_name	VERCHAR(50)	
Department	VERCHAR(50)	
salary	INT	
hire_date	DATE	

Ans:- Create a New Database and Employees Table

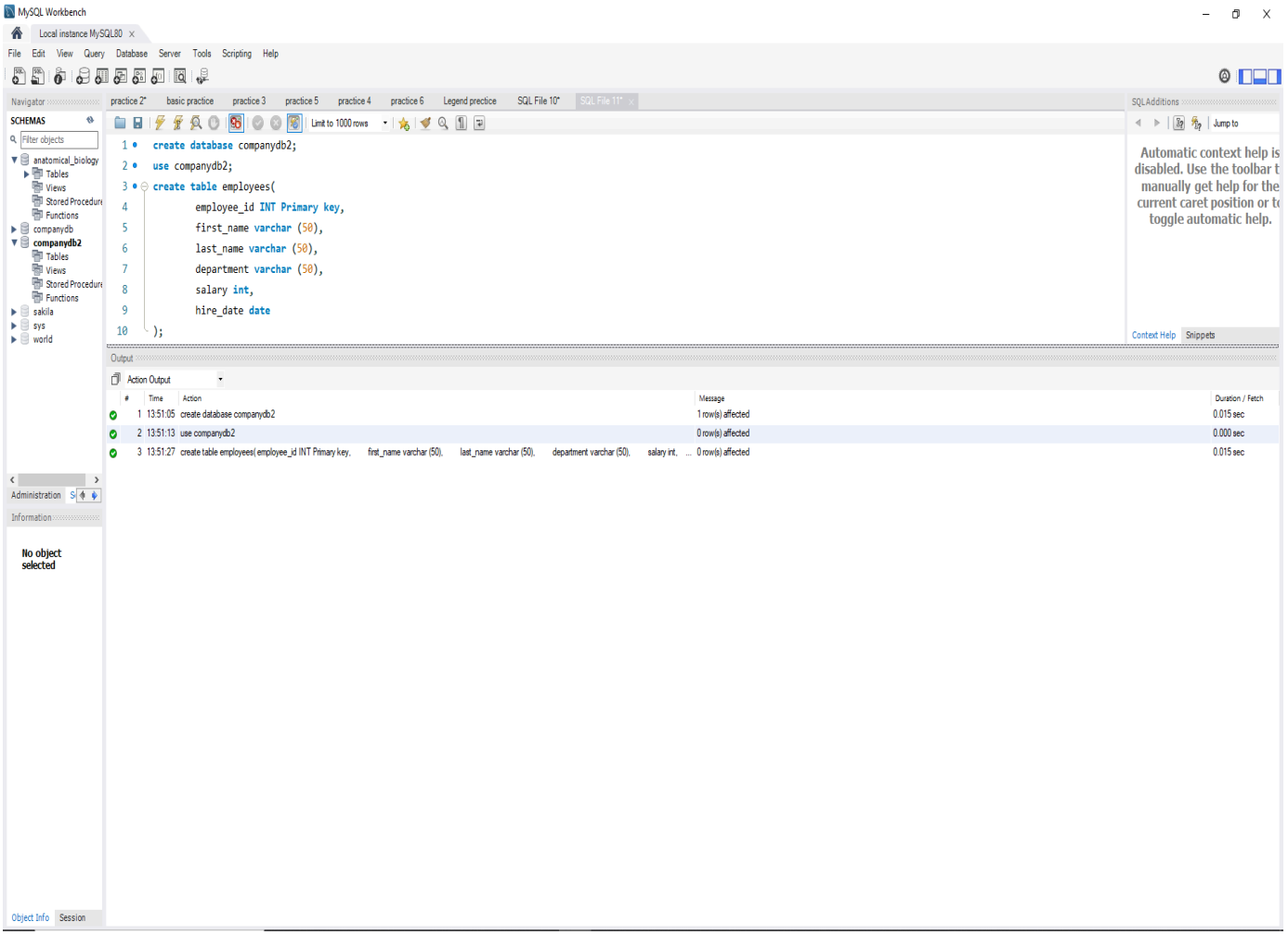
SQL Code create database companydb2;

use companydb2;

```
create table employees(
    employee_id INT Primary key,
    first_name varchar (50),
```

```
    last_name varchar (50),
    department varchar (50),
    salary int,
    hire_date date
);
```

Output Screenshot



Q2. Insert Data into Employees Table

Task: Insert the following sample records into the employees table.

Employee_id	first_name	last_name	Department	salary	hire_date

101	Amit	Sharma	HR	50000	2020-01-15
102	Riya	Kapoor	Sales	75000	2019-03-22
103	Raj	Mehta	IT	90000	2018-07-11
104	Neha	Verma	IT	85000	2021-09-01
105	Arjun	Singh	Finance	60000	2022-02-10

Ans:- insert into employees(

employee_id, first_name, last_name, department, salary, hire_date)

values

(101, 'Amit', 'Sharma', 'HR', 50000, '2020-01-15'),

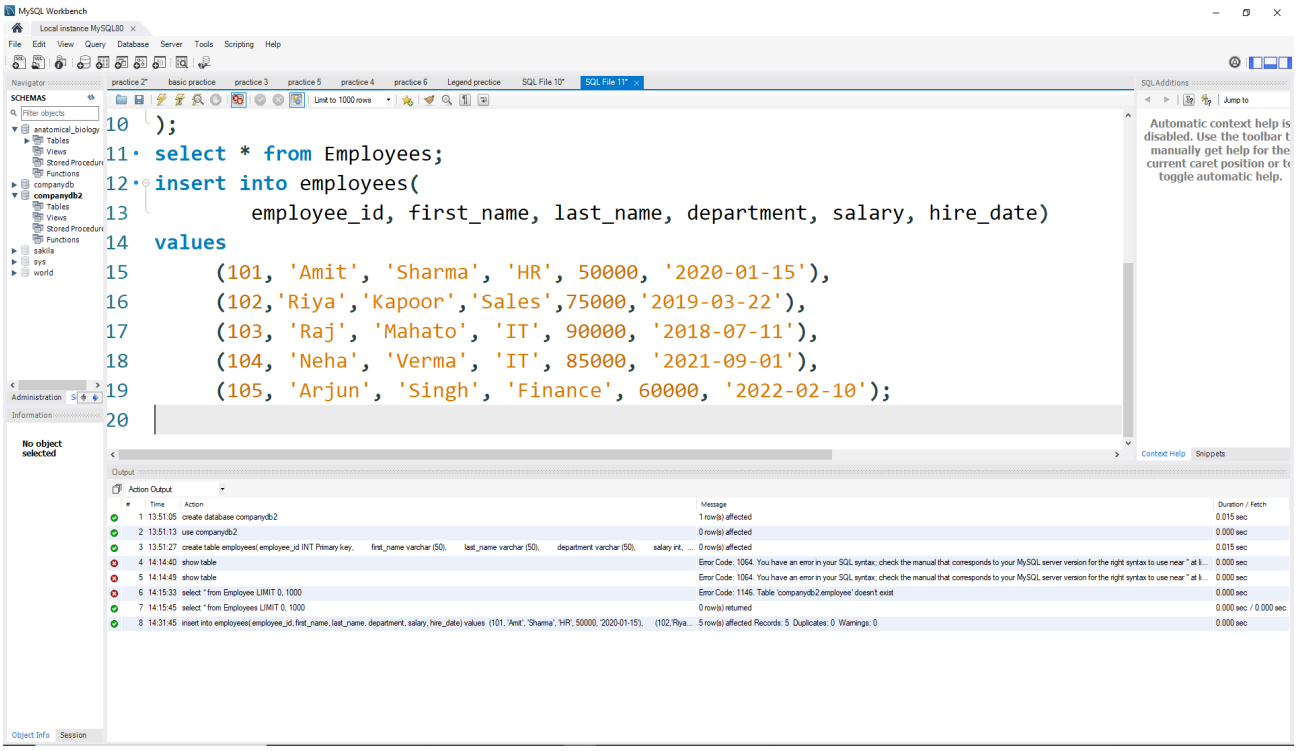
(102,'Riya','Kapoor','Sales',75000,'2019-03-22'),

(103, 'Raj', 'Mahato', 'IT', 90000, '2018-07-11'),

(104, 'Neha', 'Verma', 'IT', 85000, '2021-09-01'),

(105, 'Arjun', 'Singh', 'Finance', 60000, '2022-02-10');

Output Screenshot



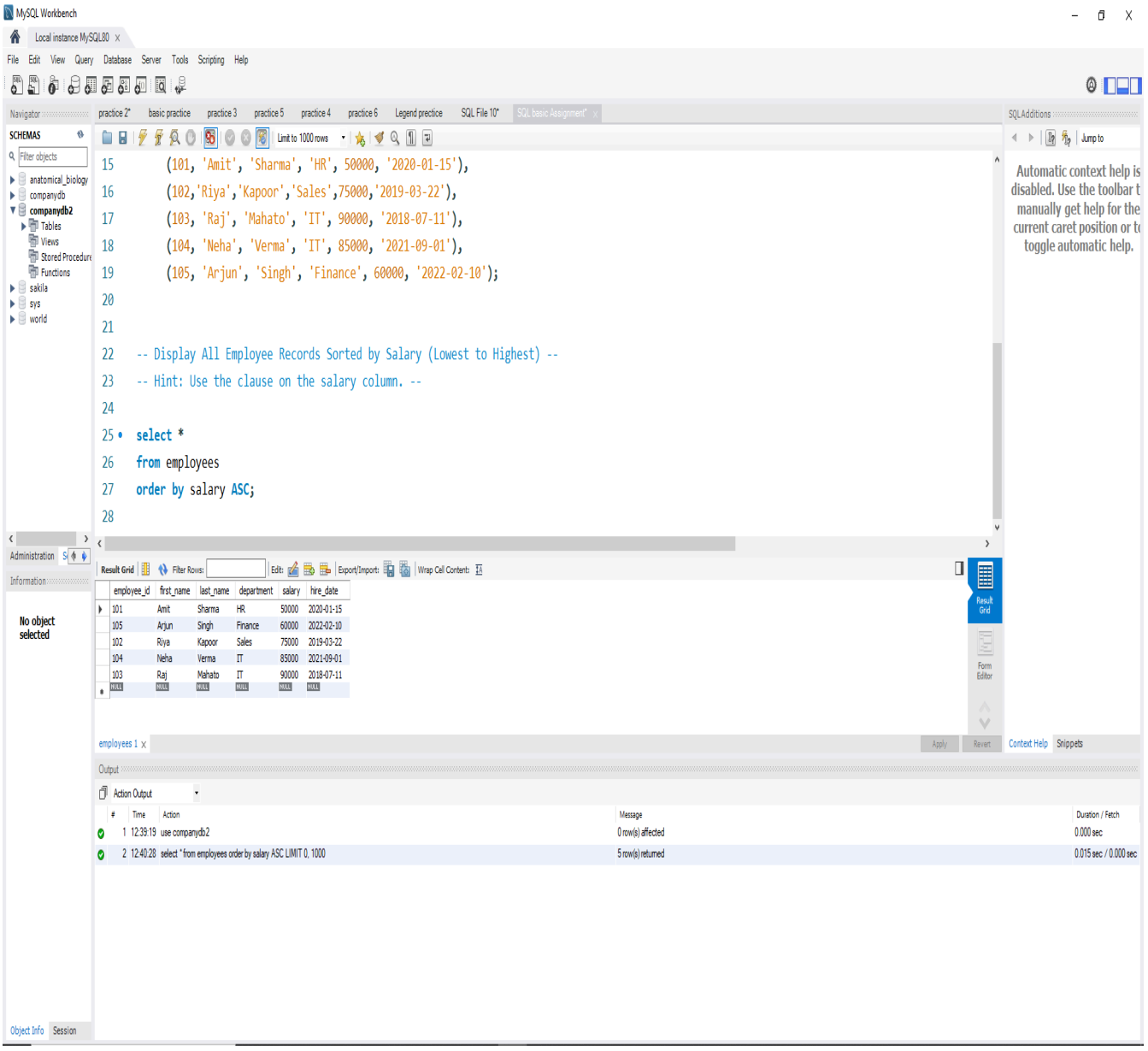
Q3. Display All Employee Records Sorted by Salary (Lowest to Highest)

Hint: Use the clause ORDER BY on the salary column.

Ans:- use companydb2

select * from employees

Order by ASC;

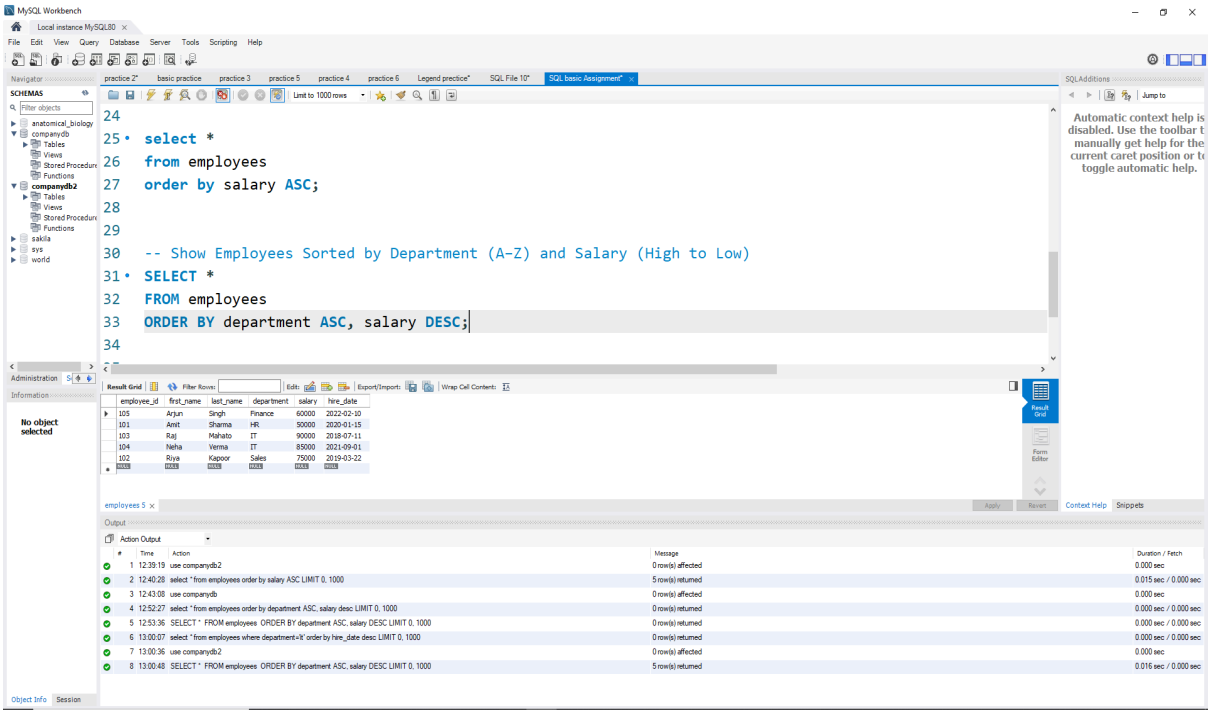


Q4. Show Employees Sorted by Department (A–Z) and Salary (Highest to Lowest).

Ans:- select *

from employees

Order by department ASC, salary DESC;

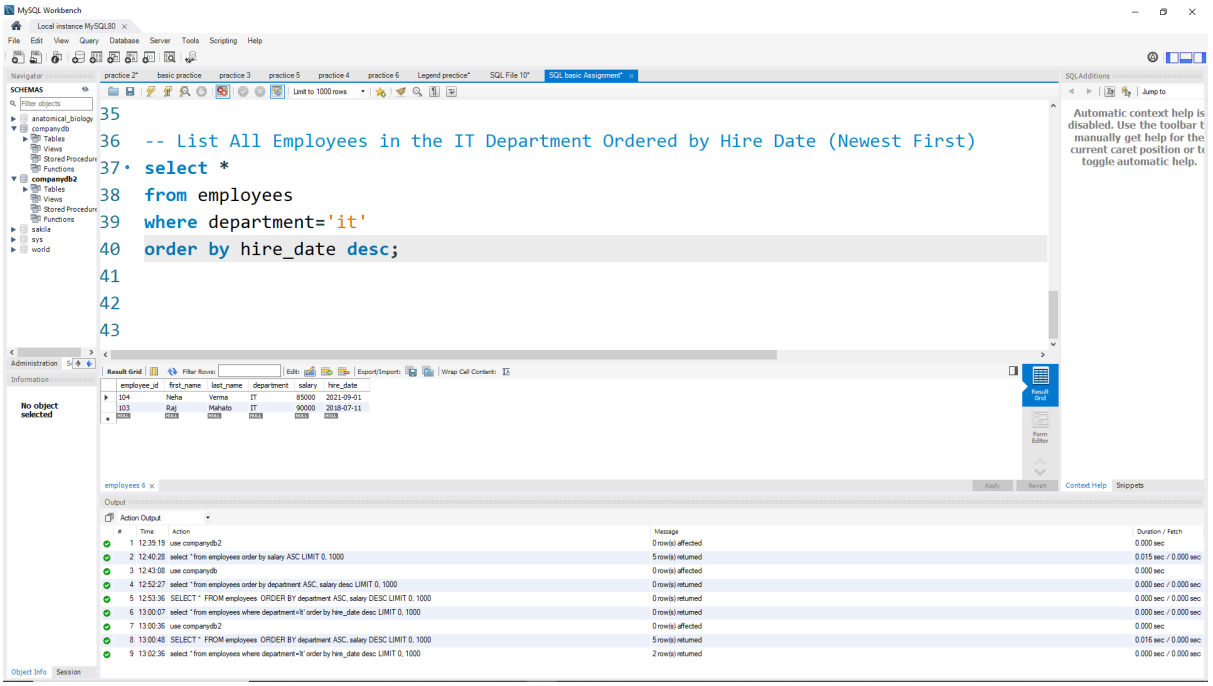


5. List All Employees in the IT Department Ordered by Hire Date (Newest First) .

Ans:- select * from employees

Where department='IT'

Order by hire_date desc;



Q6. Create and Populate Sales Table

Given data:-

Sale_id	customer_name	amount	sale_date
1	Aditi	1500	2024-08-01
2	Rohan	2200	2024-08-03
3	Aditi	3500	2024-09-05
4	Meena	2700	2024-09-15
5	Rohan	4500	2024-09-25

Ans:- create table sales(

sele_id int primary key,

customer_name varchar(50),

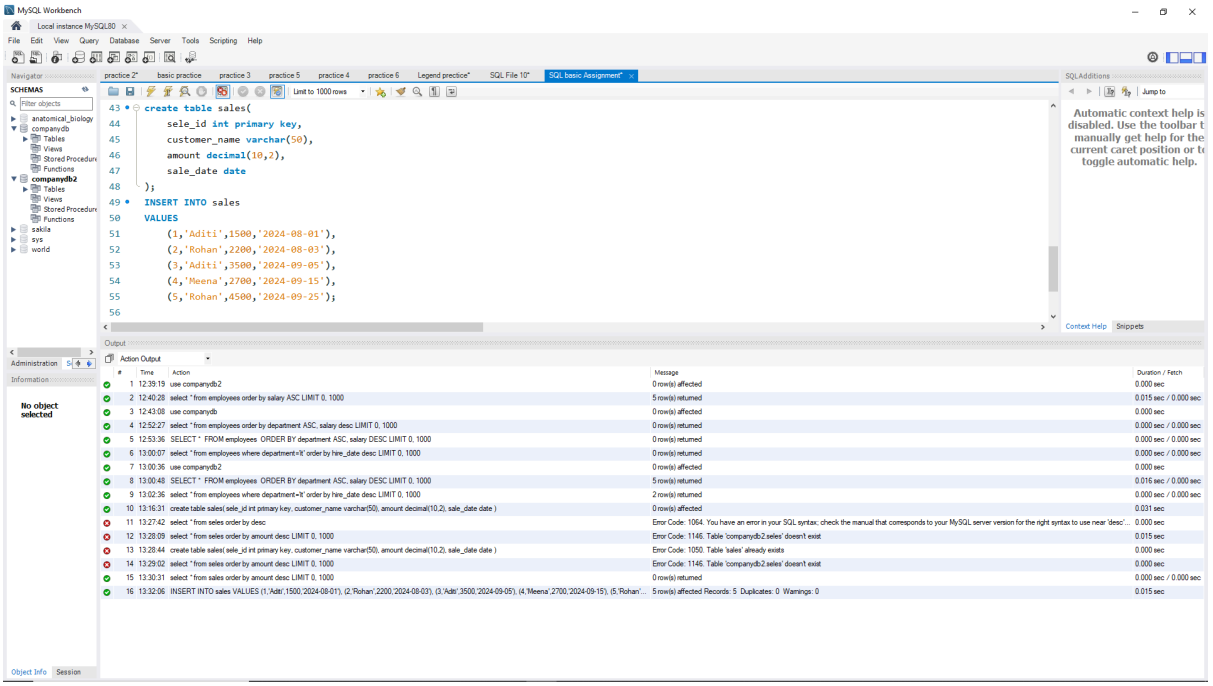
amount decimal(10,2),

sale_date date

);

```
INSERT INTO sales
VALUES
    (1,'Aditi',1500,'2024-08-01'),
    (2,'Rohan',2200,'2024-08-03'),
    (3,'Aditi',3500,'2024-09-05'),
    (4,'Meena',2700,'2024-09-15'),
    (5,'Rohan',4500,'2024-09-25');
```

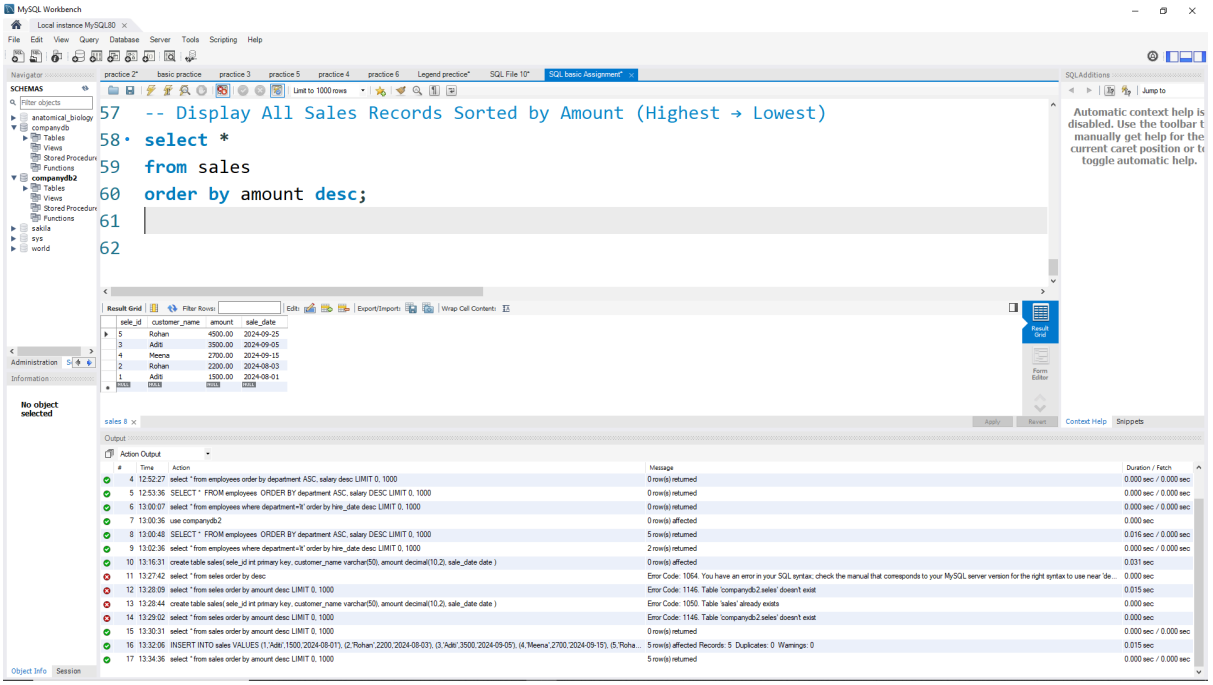
Output screenshot:-



Q7. Display All Sales Records Sorted by Amount (Highest to Lowest)

Hint: Use ORDER BY amount DESC.

Ans:-
```
select *
    from sales
    order by amount desc;
```



Q8. Show All Sales Made by Customer "Aditi"

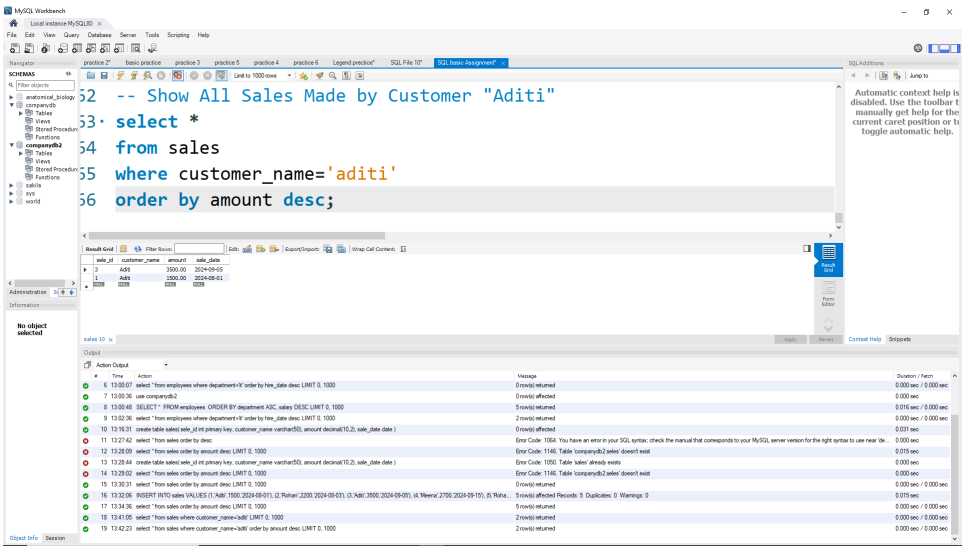
Hint: Use ORDER BY amount DESC.

Ans:- select *

from sales

where customer_name='aditi'

order by amount desc;



Q9. What is the Difference Between a Primary Key and a Foreign Key?

Ans:- Difference Between Primary Key and Foreign Key

Primary Key	Foreign Key
● Uniquely identifies each record in a table.	● Creates a relationship between two tables.
● Cannot contain NULL values.	● Can contain NULL values (depending on design).
● Must be unique.	● Can contain duplicate values.
● Only one Primary Key is allowed per table.	● A table can have multiple Foreign Keys.

Q10. What Are Constraints in SQL and Why Are They Used?

Ans:- Constraints are rules applied to table columns to ensure the accuracy and integrity of data.

Common SQL Constraints

- PRIMARY KEY
- FOREIGN KEY
- NOT NULL
- UNIQUE
- CHECK
- DEFAULT

Conclusion

This assignment demonstrates the creation of databases and tables, inserting records, retrieving data using SQL queries, sorting records, filtering data using the WHERE clause, and understanding SQL keys and constraints.

Thank You